

# Machine Learning

## *Assessed Coursework 1*

This assignment must be submitted by May 27<sup>th</sup>, 2025 at 10:00 am.  
Late submissions will be penalized by 10% **per hour of delay**.

This coursework is assessed and mandatory  
and is worth 22% of your total final grade for this course.

### Instructions

#### Identifier

Please use the random number of 6 digits you choose for the lab and include it, as a comment, in the first cell of each the notebook files you will submit – for example, “#723923”. Make sure that you keep a copy of that number as it will be used to provide the feedback (please avoid trivial numbers, such as 000000 or 123456. Also please avoid numbers starting with zero).

#### Submission

Submit your notebook files (compressed in zip or rar file) through EClass (name it “723923.rar” following your selected ID). The file you submit cannot be read by any other students. You can replace your submission as many times as you like by resubmitting it, although only the last version sent is kept.

*If you have last-minute problems with your EClass submission, email your assignment as an attachment to [alberto.paccanaro@fgv.br](mailto:alberto.paccanaro@fgv.br) with the subject "URGENT – Coursework 1 SUBMISSION". In the body of the message, explain the reason for not submitting it through EClass.*

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| <p><b>All the work you submit should be solely your own work.<br/>Coursework submissions will be checked for this.</b></p> |
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**You must add your code to the given notebook file for each exercise.**

**Note that some imports are already included in the notebook files and you are not allowed to include further ones.**

**All functions you need must be included in the notebook file of each exercise.**

### **EXERCISE 1 (Value: 8 marks)**

In this exercise you will implement the EM algorithm for mixture of Gaussians. The exercise is composed of two parts. In the first part, you will use data in 2D contained in file exercise1\_2.mat. Here you will be able to take advantage of the fact that you can visualize the data. In the second part, you will use data in 10D contained in file exercise1\_10.mat.

#### **Part 1**

File "exercise1\_2.mat" contains two variables, C0 and C1, corresponding to points in classes  $C_0$  and  $C_1$  respectively.

You will implement the EM algorithm for mixture of Gaussians and you will apply it to the problem of classifying a 2D point into classes  $C_0$  and  $C_1$ .

When running your code, every few steps (you decide how many) your code will plot the training points and the mean of each gaussian as a scatter plot (use dots for training points and crosses of a different colour for the means of the gaussians). At the end of the training, you will plot another figure, showing the log-likelihood during the training. All your plots must include axis labels, titles and legends.

Your algorithm will stop when absolute difference of the log-likelihood between 2 consecutive iterations is smaller than  $10^{-6}$ .

Having learned a model for each of the classes, you will evaluate your predictions on the test set using accuracy, precision and recall.

Points will be given for correctly deciding on the model hyperparameters using a validation set.

*Tip: when implementing the EM algorithm, it helps to initialize the covariances to the covariance of a subset of your data (rather than to random values).*

#### **Part 2**

File "exercise1\_5.mat" contains 3 variables, C0, C1, C2, corresponding to points in classes  $C_0$ ,  $C_1$  and  $C_2$  respectively.

You will implement the EM algorithm for mixture of Gaussians and you will apply it to the problem of classifying a 5D point into classes  $C_0$ ,  $C_1$  and  $C_2$ .

At the end of the training, you will plot a figure showing the log-likelihood during the training. Your plot must include axis labels and title.

Your algorithm will stop when absolute difference of the log-likelihood between 2 consecutive iterations is smaller than  $10^{-6}$ .

Having learned a model for each of the classes, you will evaluate your predictions on the test set using accuracy, precision and recall per class.

Points will be given for choosing the model hyperparameters using a validation set.

*Tip: when implementing the EM algorithm, it helps to initialize the covariances to the covariance of a subset of your data (rather than to random values).*

## EXERCISE 2 (Value: 7 marks)

The data for this exercise is contained in file “diabetes.csv” containing 768 points in 9 dimensions. This dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases and its objective is to diagnostically predict whether or not a patient has diabetes, based on certain diagnostic measurements included in the dataset. Several constraints were placed on the selection of these instances from a larger database. In particular, all patients here are females at least 21 years old of Pima Indian heritage. Each point refers to a patient, which is described by 8 attributes, all of them in the first 8 columns. A patient diagnosed with diabetes is labeled as 1 in the last column, and 0 otherwise.

The attributes of the patients include:

1. Pregnancies: number of times pregnant.
2. Glucose.
3. Blood Pressure.
4. Skin Thickness.
5. Insulin.
6. BMI: Body mass index.
7. Diabetes Pedigree Function.
8. Age.

In this exercise, you will build a Naïve Bayes classifier to predict if a patient has diabetes using its attributes. You will evaluate your predictions on the test set using accuracy, precision and recall.

Points will be given for choosing the model hyperparameters using a validation set.

## EXERCISE 3 (Value: 7 marks)

The data for this exercise is contained in file “Concrete\_Data.csv” containing 1030 datapoints in 9 columns. The concrete compressive strength is a highly nonlinear function of age and ingredients. These ingredients include cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, and fine aggregate. This dataset contains the actual concrete compressive strength (MPa) for a given mixture at a specific age (days) as it was determined from laboratory. Data is in raw form (not scaled), each row represents a specific mixture of concrete: the first 7 columns contain values for the concrete ingredients, the 8<sup>th</sup> column contains the concrete age, and the 9<sup>th</sup> column contains the target, that is the concrete compressive strength.

In this exercise, you will build and train a neural network using numpy to predict the concrete compressive strength given the 8 features (ingredients and age) in the dataset. You must implement the forward and backward pass of the backpropagation algorithm yourself and train your neural network using the batch gradient descent learning strategy. You need to regularize the weights of your neural network using weight decay.

Your implementation will plot the training error as a function of the iterations. The algorithm will stop when the relative improvement in the error function between two successive iterations will be smaller than the value of a user-defined threshold  $\tau$ .

Having learned a model for each of the classes, you will evaluate your predictions on the test set using root mean square error.

Points will be given for choosing the model hyperparameters using a validation set.